Ratan Tata Maharashtra State Skill University Pune

Department of Computer Technology

Project Report on

"Guest Lecture & Visit Management System"

Subject-: Capstone project

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CERTIFICATE

This is to certify that Mr. / Ms._____Roll No_____of Sixth Semester degree in Computer Technology of Institute **Ratan Tata Maharashtra state skill university Pune** has completed the Final project Report on topic **(Guest Lecture & Visitor Management System)** in subject of Capstone Project Planning for academic year 2025 to 2026 as prescribed in the curriculum.

Place: -…… Eligibility No: - .....

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Chapter No 01

ABSTRACT

The **Guest Lecture & Visit Management System (GLVMS)** is a web-based application developed to automate and streamline the process of managing campus visitors and organizing guest lectures in an educational institution. The system replaces traditional manual entry registers and unstructured coordination methods with a secure, efficient, and digital workflow. It addresses common challenges such as delays in visitor verification, lack of proper tracking, and inefficient management of guest lecture arrangements.

The primary objective of this project is to develop a centralized platform that enables visitors to request entry, administrators to manage approvals, and security personnel to verify access through a QR-based system. Visitors can register by filling out an online form with details such as name, purpose of visit, department, and identity proof. Once approved by the administrator, the system generates a unique QR code and sends it to the visitor as a digital entry pass. Security personnel can scan this QR code at the entry point to validate and record entry and exit details in real time.

The system is developed using modern web technologies, with the frontend built using React.js and Tailwind CSS to ensure a responsive and user-friendly interface. The backend is implemented using Node.js and Express.js, and MySQL is used as the database, managed through Sequelize ORM for efficient data handling and scalability. The modular architecture of the system separates functionalities such as authentication, visitor management, lecture management, and notification services, making the application maintainable and extensible.

In addition to visitor management, the system provides a dedicated module for managing guest lectures. Administrators can schedule lectures, assign coordinators, and manage venues. Staff members can track preparation tasks through checklists and record expenses related to the event. The system also allows participants to submit feedback after attending lectures, helping improve the quality of future events.

Key outcomes of the project include:

- A QR-based secure and contactless entry system for visitors
- Efficient management of guest lectures with checklist and expense tracking
- A centralized admin dashboard for monitoring and control
- Role-based access for Admin, Security, Staff, and Visitors
- Reliable data storage and management using MySQL and Sequelize ORM

In conclusion, the Guest Lecture & Visit Management System provides a comprehensive digital solution for managing campus visits and academic events. It enhances operational efficiency, reduces manual errors, and ensures real-time monitoring and control. The system is scalable and can be further extended with features such as advanced analytics, biometric verification, and integration with institutional systems, making it suitable for modern smart campus environments.

Chapter No 02
INTRODUCTION

2.1 Background of the Project

In many educational institutions, the process of managing campus visitors and organizing guest lectures is still handled manually. Visitor entries are recorded in physical registers, and approvals are often communicated through informal channels. This traditional approach leads to inefficiencies such as long waiting times, lack of proper record management, and increased chances of errors. Similarly, guest lecture coordination involves multiple departments and manual tracking of tasks, making the process time-consuming and difficult to manage.

With the advancement of digital technologies, there is a growing need to replace these outdated methods with automated systems. The **Guest Lecture & Visit Management System (GLVMS)** is developed to modernize these processes by providing a centralized, digital platform that ensures accuracy, security, and efficiency.

2.2 Motivation

The motivation behind this project arises from the real-world challenges faced in managing visitor entries and academic events within institutions. Manual systems are prone to human errors, miscommunication, and lack of real-time monitoring. Security concerns also increase due to improper verification of visitors.

The need to create a smart, secure, and user-friendly system that simplifies these operations motivated the development of GLVMS. The aim is to leverage modern technologies such

QR code authentication, web-based interfaces, and automated workflows to improve overall efficiency and user experience.

2.3 Need of the System

The existing manual system has several limitations, including:

- Lack of real-time tracking of visitors
- Time-consuming entry and verification process
- Poor coordination in managing guest lectures
- High chances of data loss or duplication
- Limited transparency and accountability

To overcome these issues, a digital system is required that can:

- Automate visitor registration and approval
- Provide secure QR-based entry verification
- Maintain accurate and centralized records
- Streamline lecture management processes
- Improve communication between users

2.4 Problem Definition

The main problem addressed by this project is the inefficiency and lack of security in traditional visitor management and guest lecture coordination systems. Manual processes make it difficult to track visitor movements, ensure proper authorization, and manage event-related activities effectively.

There is a need for a system that can:

- Digitally manage visitor requests and approvals
- Provide secure and contactless entry using QR codes
- Enable efficient coordination of guest lectures
- Maintain structured data for analysis and reporting

The absence of such a system results in operational delays, reduced security, and poor management of institutional activities.

2.5 Project Scope

The scope of the **Guest Lecture & Visit Management System** includes the development of a web-based platform that supports multiple user roles such as Admin, Security, Staff, and Visitors. The system covers functionalities such as visitor registration, approval workflow, QR code generation, entry and exit logging, lecture scheduling, checklist management, expense tracking, and feedback collection.

The project is designed to be scalable and can be extended with additional features such as biometric authentication, integration with institutional databases, advanced analytics, and mobile application support. It is suitable for deployment in colleges, universities, and similar organizational environments.

Chapter No 03
Requirement Analysis

3.1 Requirement Analysis

Requirement Analysis is the process of identifying and defining the functionalities and constraints of the system. It helps in understanding what the system should do (functional requirements) and how the system should perform (non-functional requirements).

- **Functional Requirements**

These define the core operations and features of the system:

1. **User Registration and Authentication**
   - Admin, Staff, and Security users can register and log in
   - Secure authentication mechanism is implemented
2. **Visitor Management**
   - Visitors can fill out an online visit request form
   - Admin can approve or reject requests
   - Visitor data is stored in the database
3. **QR Code Generation**
   - A unique QR code is generated after approval
   - The QR code acts as a digital entry pass
4. **QR Code Verification**
   - Security personnel scan the QR code
   - System verifies authenticity
   - Entry and exit times are recorded
5. **Lecture Management**
   - Admin can create, update, and delete lecture details
   - Lectures can be assigned to staff members
6. **Checklist Management**
   - Staff can manage and update task checklists for lectures
7. **Expense Management**
   - Staff can add and manage expenses related to lectures
8. **Feedback System**
   - Visitors can submit feedback after attending lectures
9. **Dashboard**
   - Admin dashboard provides an overview of visitors, lectures, and logs

- **Non-Functional Requirements**

These define the quality and performance aspects of the system:

1. **Performance**
    - The system should provide fast response times
    - QR verification should be real-time
2. **Security**
    - User authentication must be secure
    - Sensitive data must be protected
3. **Usability**
    - The interface should be simple and user-friendly
    - The system should be accessible on multiple devices
4. **Scalability**
    - The system should support future enhancements
    - It should handle an increasing number of users
5. **Reliability**
    - The system should operate consistently without failure
    - Data loss should be prevented
6. **Maintainability**
    - The system should be easy to update and maintain
    - Code structure should be modular

3.2 Feasibility Study

A feasibility study is conducted to determine whether the project is practical and achievable from different perspectives.

◆ Technical Feasibility

- The system uses modern technologies such as React.js, Node.js, and MySQL
- QR code integration is technically achievable
- The application is web-based, ensuring cross-platform compatibility
- Required technical skills and tools are available

Conclusion:

The system is technically feasible.

◆ Economic Feasibility

- The project uses open-source technologies, reducing development cost
- Maintenance cost is low
- It is more cost-effective compared to manual systems

Conclusion:

The system is economically feasible.

◆ Operational Feasibility

- The system is easy to use with a simple interface
- Staff and security personnel can adapt quickly
- Minimal training is required
- Suitable for real-world deployment in institutions

Conclusion:

The system is operationally feasible.

Chapter 4
System Design

1. System Architecture

The **QR-Based Hyperlocal Store Ordering System** is designed using a client–server architecture and built with lightweight, scalable technologies such as **Python, Flask, SQLite, and SQLAlchemy**. The architecture ensures fast product browsing, smooth order submission, and real-time interaction between customers and store owners through any web browser.

Client-Side (Customer & Seller Interface):

• Developed using **HTML, CSS, and JavaScript** for a clean, mobile-responsive interface.
• Customers scan the QR code to instantly open the store's online page.
• Allows users to:
– Browse products
– Add items to "Ordered Items"
– Select Pickup or Delivery
– Enter phone number and address (only for delivery)
– Submit order
• Sellers can access a dashboard to:
– Add/edit/delete products
– View customer orders
– Update order status (Pending, Accepted, Ready, Delivered)
– Control store status (Open/Closed)
• All pages are served through Flask's **render_template()** for smooth frontend-backend integration.

Server-Side (Backend + Business Logic):

• Built using **Flask** for routing, session handling, and logic processing.
• Uses **SQLite + SQLAlchemy** to store sellers, products, orders, delivery info, and store settings.
• Generates a **unique store slug** and **QR code** for every new seller at registration.
• Handles all core functions:
– Product management
– Order storage & retrieval
– Delivery-charge calculation
– Store open/close verification
– Validation of customer inputs
• Ensures secure, quick, and error-free data processing.

Communication Layer:

• A REST-style POST request is sent by the client to endpoints such as:
– /submit_order (store customer order)
– /add_product (seller adds product)
– /update_status (seller changes order status)
• Backend processes user input, updates the database, and returns responses to be displayed instantly in the UI.
• JSON is used for exchanging order details and responses when needed.

Chapter N0 5

Implementation

5.1 Module Description

The **Guest Lecture & Visit Management System (GLVMS)** is divided into multiple modules to ensure a structured and maintainable system design. Each module performs a specific function and interacts with others to complete the workflow.

- **1. Authentication Module**

- Handles user registration and login
- Supports role-based access (Admin, Staff, Security)
- Ensures secure access to the system

- **2. Visitor Management Module**

- Allows visitors to submit visit requests
- Stores visitor details in the database
- Admin can approve or reject requests

- **3. QR Code Module**

- Generates a unique QR code for approved visitors
- Acts as a digital entry pass
- Ensures secure and contactless verification

- **4. Security Verification Module**

- Enables QR code scanning at the entry gate
- Verifies visitor authenticity
- Records entry and exit timestamps

- **5. Lecture Management Module**

- Allows admin to create and manage guest lectures
- Assigns staff coordinators
- Stores lecture details (topic, speaker, venue, date)

- **6. Checklist Management Module**

- Staff can track preparation tasks
- Tasks can be marked as completed or pending

- **7. Expense Management Module**

- Records expenses related to lectures
- Helps in financial tracking

- **8. Feedback Module**

- Visitors can submit feedback after attending lectures
- Helps improve future events.

5.2 Algorithm

- **Visitor Approval & QR Generation Algorithm**

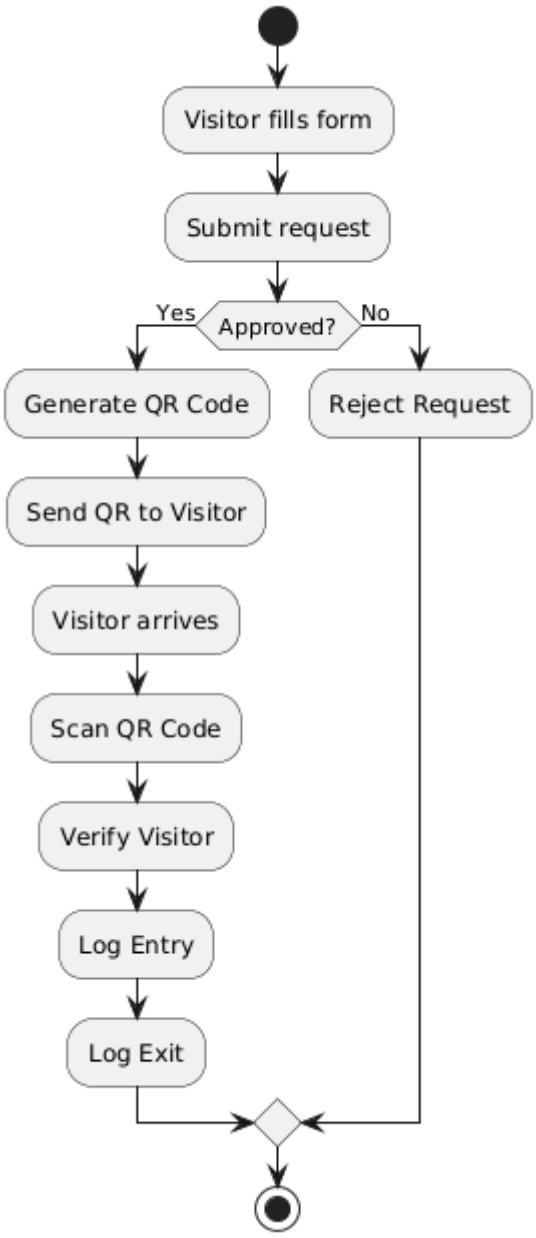
1. Start
2. Visitor fills and submits the visit request form
3. System stores request in database with status = "Pending"
4. Admin reviews the request
5. If approved:
    - Generate unique QR code
    - Update status to "Approved"
    - Send QR code to visitor
6. Else:
    - Update status to "Rejected"
7. End

- **QR Verification Algorithm**

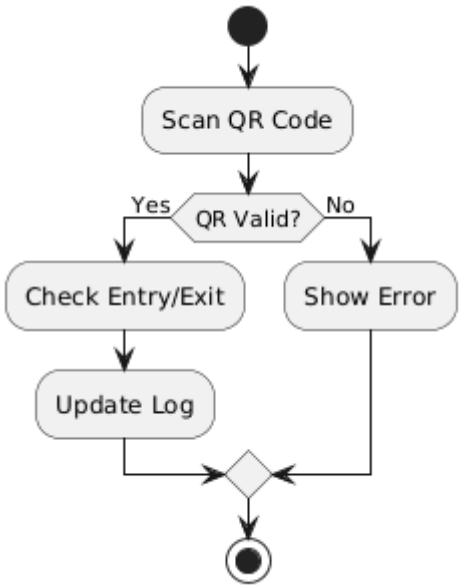
1. Start
2. Security scans QR code
3. System checks QR validity in database
4. If valid:
    - Record entry time (if first scan)
    - Record exit time (if second scan)
5. Else:
    - Show invalid QR message
6. End

5.3 Flowcharts:

Visitor Flowchart:



QR Verification Flowchart:



5.4 Screenshots of System:

- Landing Page
- Signup / Login Page
- Visitor Registration Form
- Admin Dashboard
- QR Code Display
- QR Scanner Page
- Lecture Management Page
- Checklist & Expense Page

Chapter N0 6
Testing

6.1 Test Plan

The test plan defines the strategy and approach used to validate the functionality and performance of the **Guest Lecture & Visit Management System (GLVMS)**. The main objective of testing is to ensure that all modules work correctly, meet the specified requirements, and are free from errors.

◆ Testing Objectives

- To verify that all system functionalities work as expected
- To ensure secure and accurate QR-based verification
- To check system performance under normal usage
- To validate data storage and retrieval from the database

◆ Testing Scope

The following modules are tested:

- Authentication Module
- Visitor Management Module
- QR Code Generation & Verification Module
- Lecture Management Module
- Checklist & Expense Module

◆ Testing Approach

- Manual testing is performed for UI and workflows
- Functional testing is used to validate features
- End-to-end testing ensures complete system flow

6.2 Test Cases Table:

Test Case ID	Module	Input	Expected Output	Result
TC1	Login	Valid credentials	User logged in successfully	Pass
TC2	Login	Invalid credentials	Error message displayed	Pass
TC3	Visitor Form	Valid data	Request submitted successfully	Pass
TC4	Visitor Approval	Approve request	QR code generated	Pass
TC5	QR Scan	Valid QR	Entry logged successfully	Pass
TC6	QR Scan	Invalid QR	Error message displayed	Pass
TC7	Lecture Module	Add lecture	Lecture added successfully	Pass
TC8	Checklist	Update status	Status updated	Pass
TC9	Expense Module	Add expense	Expense recorded	Pass
TC10	Feedback	Submit feedback	Feedback saved	Pass

6.3 Types of Testing

- **1. Unit Testing**

  Unit testing is performed to test individual components or modules of the system independently. Each function or module is tested to ensure it works correctly.

  Example:

  - Testing login function
  - Testing QR code generation

- **2. Integration Testing**

  Integration testing ensures that different modules of the system work together correctly. It checks data flow between components.

  Example:

  - Visitor request → Admin approval → QR generation
  - QR scan → Database update

- **3. System Testing**

  System testing is performed on the complete system to verify that it meets all functional and non-functional requirements.

  Example:

  - Complete visitor flow from registration to exit
  - Lecture creation to checklist completion

- **4. User Acceptance Testing (UAT)**

  User Acceptance Testing is performed by end users to ensure that the system meets real-world requirements and is easy to use.

  Example:

  - Admin testing dashboard usability
  - Security testing QR scanning process
  - Staff testing checklist and expense modules

Chapter N0 7

CONCLUSION

7.1 Summary of Work

The **Guest Lecture & Visit Management System (GLVMS)** was successfully designed and developed as a web-based application to automate visitor entry management and guest lecture coordination within an institution. The system replaces traditional manual processes with a digital, structured, and efficient workflow.

The project includes key functionalities such as online visitor registration, admin approval system, QR code generation for secure entry, real-time entry and exit logging, and a dedicated module for managing guest lectures. Additional features like checklist management, expense tracking, and feedback collection enhance the overall usability and effectiveness of the system.

The system was developed using modern technologies including React.js for the frontend, Node.js and Express.js for the backend, and MySQL with Sequelize ORM for database management. A modular approach was followed to ensure scalability, maintainability, and smooth integration between different components.

7.2 Objectives Achieved

The project successfully achieved its intended objectives, including:

- Development of a centralized platform for visitor and lecture management
- Implementation of a secure QR-based entry system
- Automation of visitor approval and verification processes
- Efficient management of guest lectures, including scheduling and coordination
- Real-time tracking of visitor entry and exit
- Role-based access for Admin, Security, Staff, and Visitors
- Improved data accuracy and reduction in manual errors

7.3 Key Learnings

During the development of this project, several important technical and practical learnings were gained:

- Understanding of full-stack web development using React.js and Node.js
- Implementation of RESTful APIs and backend logic using Express.js
- Database design and management using MySQL and Sequelize ORM
- Integration of QR code technology for real-world applications
- Importance of modular system design for scalability and maintainability
- Experience in handling real-world problems and designing efficient solutions
- Improvement in debugging, testing, and problem-solving skills

Chapter 8:
References

The following references were used during the development of the Guest Lecture & Visit Management System (GLVMS):

- **Books**

1. ***Web Technologies: HTML, CSS, JavaScript*** **– Kogent Learning Solutions**
2. ***Node.js Web Development*** **– David Herron**
3. ***Learning React*** **– Alex Banks & Eve Porcello**
4. ***Database System Concepts*** **– Abraham Silberschatz**

- **Web Resources**

1. **React Official Documentation – https://react.dev**
2. **Node.js Official Documentation – https://nodejs.org**
3. **Express.js Documentation – https://expressjs.com**
4. **MySQL Documentation – https://dev.mysql.com/doc**
5. **Sequelize ORM Documentation – https://sequelize.org**
6. **Tailwind CSS Documentation – https://tailwindcss.com**
7. **QR Code Generator Library – https://www.npmjs.com/package/qrcode**

- **Tools & Platforms**

1. **Visual Studio Code (Code Editor)**
2. **Postman (API Testing Tool)**
3. **GitHub (Version Control)**
4. **MySQL Server (Database Management)**
5. **Browser Developer Tools (Debugging)**

- **Online Tutorials & Learning Platforms**

1. **YouTube (Various educational channels)**
2. **W3Schools – https://www.w3schools.com**
3. **GeeksforGeeks – https://www.geeksforgeeks.org**
4. **Stack Overflow – https://stackoverflow.com**